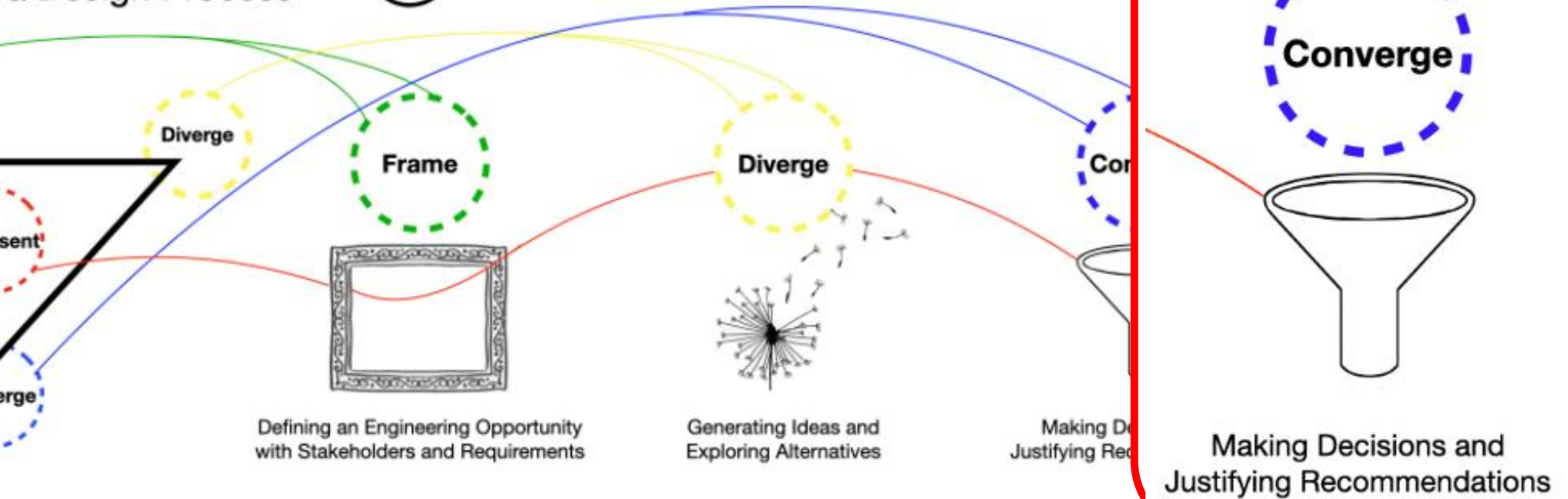


f Praxis

a Foundation:
a Design Process

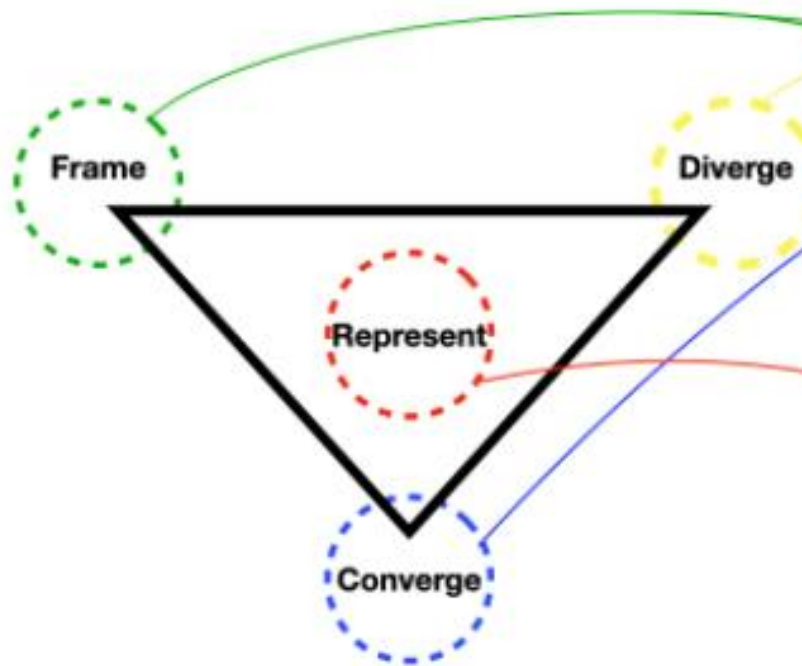
② The Strands: Developing each Step + Skills



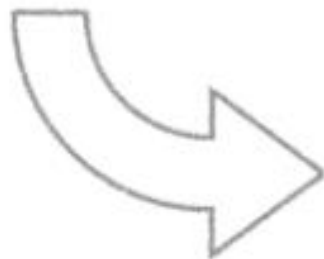
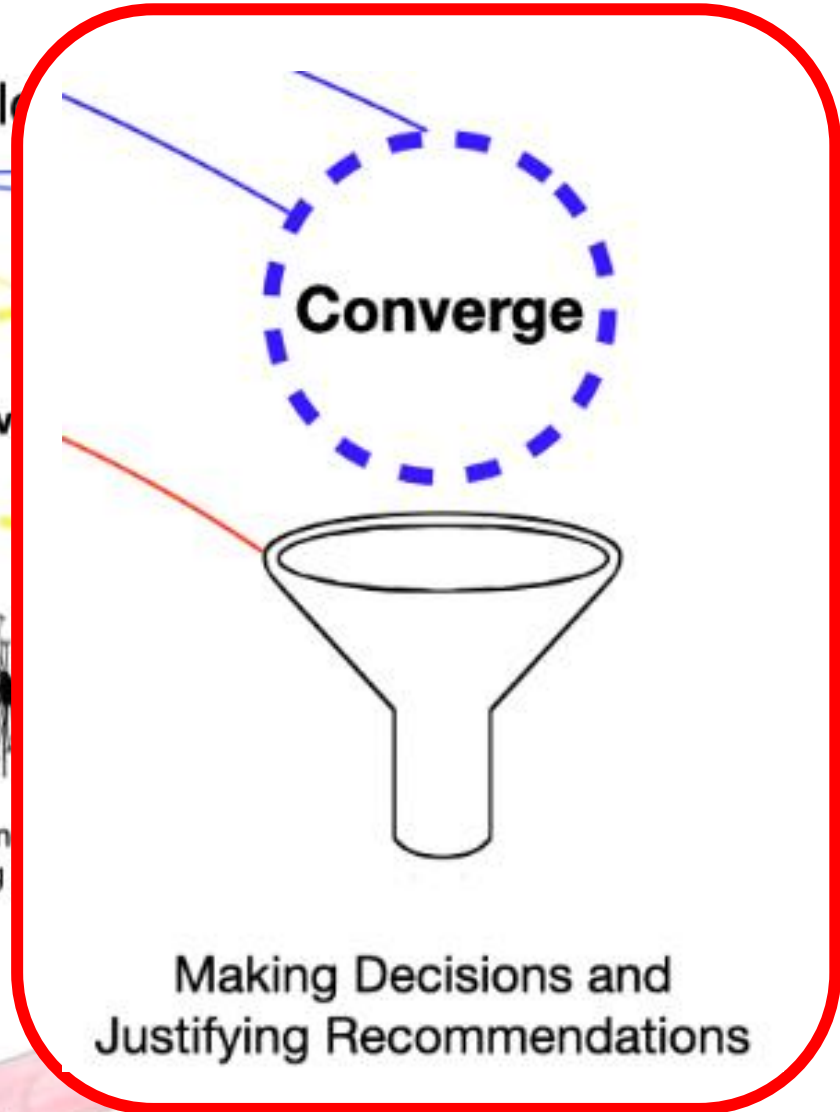
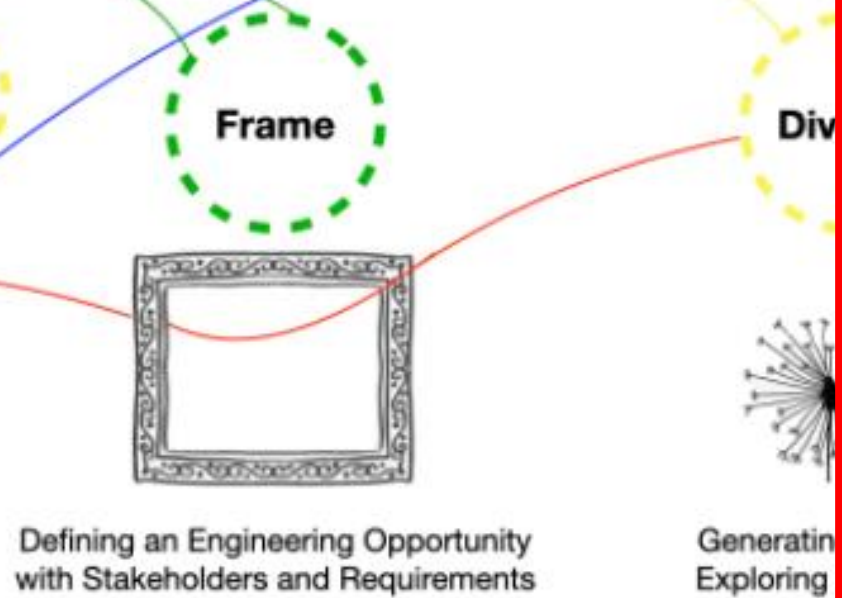
③ The Final: Putting It All Together

Visual Structure of Praxis

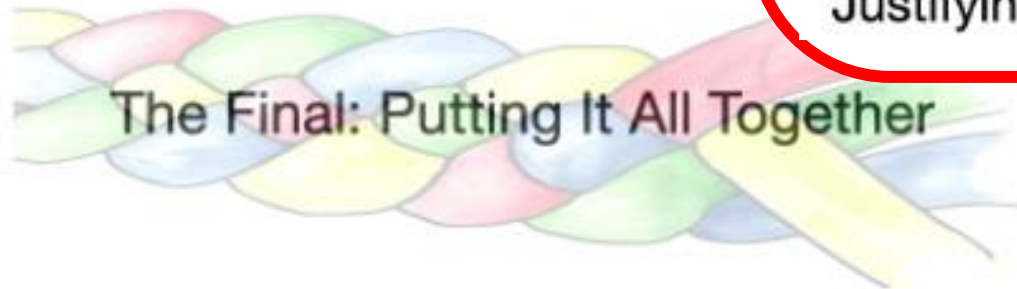
① Building a Foundation:
Learning a Design Process



② The Strands: Develop



③ The Final: Putting It All Together



Design Report

Due: 2025-11-29 at 2200 **Weight:** 20% **Submitter:** Team

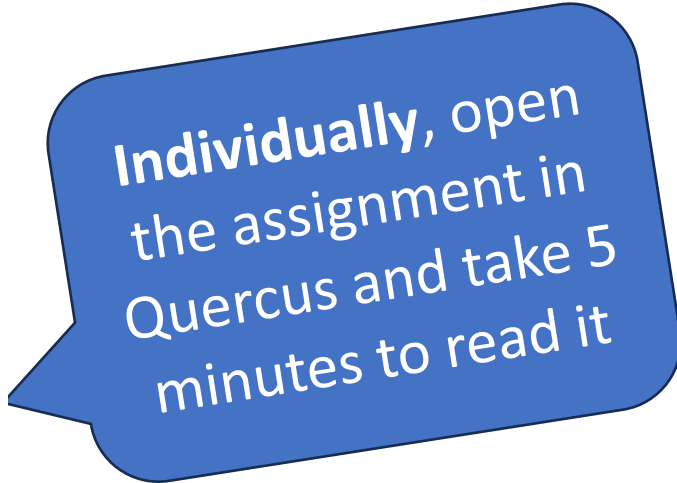
Assignment overview

The design report represents your design work in its entirety to a reader unfamiliar with your opportunity framing and design work. So, the design report documents both the design concept you are recommending, and the key parts of the process to arrive at that concept, with an emphasis on your recommendation. The design report answers the question of whether the current design is a verifiable and verified solution to the opportunity presented in the design brief. More specifically, the report aims to answer three key questions:

1. What your recommended design concept, and why is it best current design concept? The word “best” implies comparison, so you should be able to justify the current concept relative to other concepts (and reference designs).
2. What key decisions have determined the design? As you make specific decisions, you need to be sure you understand why you made those decisions and establish the research, testing, or conceptual logic to justify your work.
3. Has the current design concept met sufficient testing to be justifiable as a solution? This question requires you to be able to document your requirements, how they were met, and the kinds of testing you have done.

To accomplish this your team should document your opportunity as reframed, key alternatives from your diverging process, the evaluation of the alternatives using appropriate research and testing, and the recommended design you converged on, though not necessarily in that order.

The design report as a genre in engineering writing does not normally assume the existence of an associated design brief, so it should be written to stand alone. For this assignment, your team will have modified the framing from a design brief, and you should focus your design report on the final framing your team used, rather than on the process of reframing from the design brief.



Individually, open the assignment in Quercus and take 5 minutes to read it

Activity 01 (a): Pairwise Prep [15 min]

Use the worksheet to prepare to champion your selected concept. Consider both arguments for the concept and against it. Support your claims with concrete evidence from your concept development and include relevant justification.

When you're done, read your teammates' arguments.

FOR

Concept: _____

Arguments for concept

Claim	Claim	Claim
Evidence	Evidence	Evidence
Justification	Justification	Justification

AGAINST

Arguments against concept

Claim	Claim	Claim
Evidence	Evidence	Evidence
Justification	Justification	Justification

Activity 02 (b): Pairwise Comparison (Adversarial Version) [30 min]

① Make a Grid

② Designate the following

1. An “adversary” for each concept in the pair,
2. A “recorder”
3. 1 moderator / point counter

	A	B	C	D
A	-	0	1	1
B	1	-	1	0
C	0	0	-	1
D	0	1	0	-

③ Run a pairwise comparison with of the alternatives

> # of team members + 1

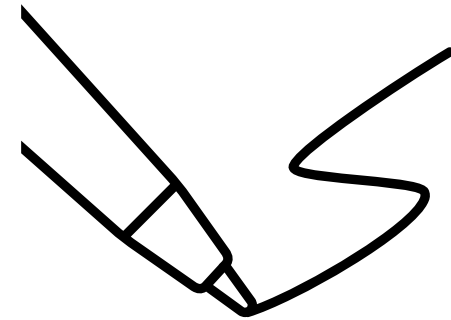
- Each adversary argues against their design (to steelman an argument “for” the other one). Arguments should reference requirements and evaluation criteria
- Recorder notes “attributes” each design does or does not achieve. Also note requirements or evaluation criteria that are referenced
- Moderator chooses the order of adversary and ensures each adversary speaks without interruption. Consider timing the rounds.

Note if some requirements or ECs get more attention

④ When you’re done discuss if you need to revise your requirements/E.C.s to account for hidden attributes that emerged?

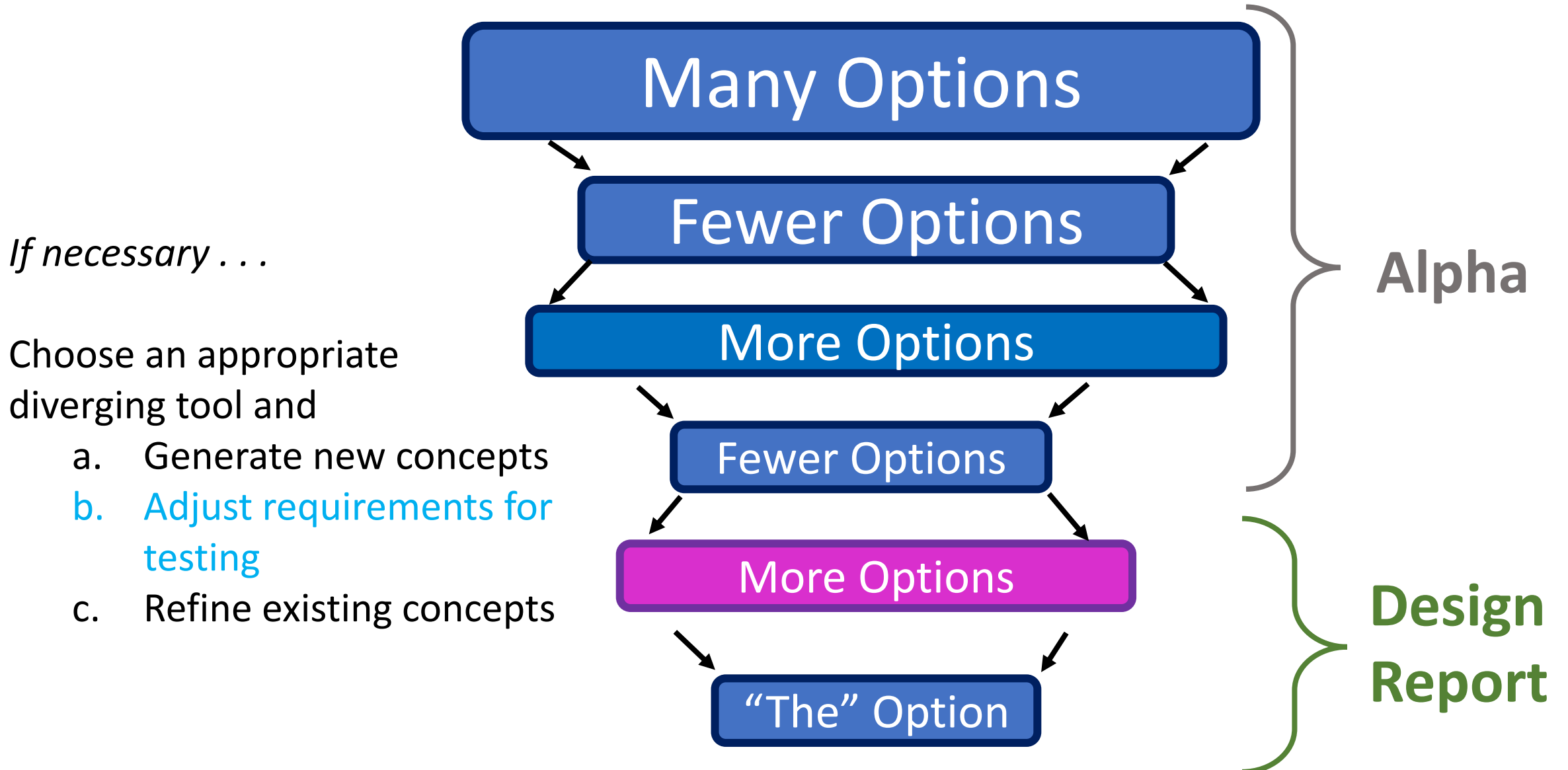
Activity 02: Writing Reflection Prompt [10 minutes]

What did you learn about
your design concepts from
the pairwise comparison?



How will this help you improve them?

Activity 03: (Re) Diverge [15 min]



How do you
arrive at “THE
option”?

Verification

- Prototyping and Research
- Proxy Testing
- Measuring and Comparing

Activity 04: Planning Prototypes and Tests [30 minutes]

1. Discuss: which requirements or ECs can be addressed by research? which require testing?
2. Identify 1 or 2 critical metrics to be used for testing. Refine as necessary.
3. Research possible “ideal” tests (standards / handbooks)
4. Develop a “proxy” test if necessary.
5. Make a plan for running your test in next week’s studio (or the one after)
 - What resources will you need for your test
 - What concept prototypes will you need to develop?
 - Where to acquire materials?



Do Your Safety Training!!!



Activity 05: Measurements Matrix and Research [15 minutes]

Acquire values for metrics from secondary sources (*and testing*)



① Evaluation Criteria

Measurement Matrix	② Alternatives			
Metrics				
Volume (L)	6.4	2.3	16	10.5
Weight (kg)	2.1	3	2.2	5.1
Comfort (Max pressure, Pa)	70k	121k	65k	72k
Water Resistance (mm)	15k	4.2	③ Measurements (Made using the measurement process, research, calculations)	



using the weight of the book to keep consistent force

Begin to fill in your measurements matrix

At Minimum Before Next Studio:

- Select 1 or 2 critical metrics
- Plan proxy tests to gather data
- Acquire materials you will need to run your tests
- Acquire materials you will need to build prototypes
- Begin researching values for other metrics